

# GURMOHIT SINGH

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## EDUCATION

The University of Texas at Austin - GPA: 3.95

*Bachelor of Science, Computer Science*

Aug 2024 – May 2028 (Expected)

Lone Star Community College - GPA: 3.98

*Associate of Science*

Aug 2021 – May 2024

## SKILLS

**Programming Languages:** Python, C, C++, Java, TypeScript, JavaScript, C#, ARM Assembly

**ML/AI:** pandas, NumPy, scikit-learn, LightGBM, Matplotlib, feature engineering, cross-validation, hyperparameter tuning, reinforcement learning, Bayesian networks, neural networks, Ollama, OpenClaw, VQOL

**Systems/Tools:** Git, Linux, Bash, PowerShell, Qiskit, IBM Quantum, PySide6, Excel, GPU-Z, Tailscale, Discord API

**Web Development:** React, Next.js, Vite, Tailwind CSS, HTML/CSS, Vercel, Cloudflare Pages

**Coursework:** Machine Learning, Artificial Intelligence, Operating Systems, Computer Architecture, Algorithms, Linear Algebra, Energy-Efficient Computing, Quantum Computing, Programming Languages

## EXPERIENCE

**WPSPIN LLC** – Houston, TX

*Web Development Intern*

Jun – Aug 2026

- Developed static and WordPress-based websites for local business clients, translating design/content requirements into responsive, production-ready pages.
- Leveraged agentic workflows to boost productivity, reducing turnaround time by ~50% and improving codebase maintainability.
- Used HTML, CSS, and JavaScript tooling to improve page structure, mobile responsiveness, and maintainability across client sites.

**Applied Research Laboratories, UT Austin** – Austin, TX

*Quantum Optics Research Intern*

Jun – Aug 2025

- Applied quantum computing algorithms and optical lab workflows through a summer quantum optics research program.
- Implemented Quantum Teleportation and Entanglement Swapping experiments, connecting theoretical quantum information concepts to physical lab demonstrations.
- Demonstrated violation of the Bell-CHSH Inequality, reinforcing experimental concepts in entanglement, measurement, and non-classical correlations.

**Tandoorish LLC** – Houston, TX

*IT Services Manager*

Jun 2024 – Present

- Managed business-critical technology systems including POS workflows, online ordering, website updates, email/account access, printers, and Wi-Fi/network troubleshooting.
- Served as the first point of contact for staff technology issues, resolving platform access, device, ordering, connectivity, and vendor-support problems during daily operations.
- Used SEO and Google Analytics insights to improve online visibility, contributing to a 200%+ increase in site traffic since takeover.

## PROJECTS

[ARM Visualizer](#) | *TypeScript, React, Vite, Vitest, Playwright*

Dec 2025 – Present

- Built an interactive AArch64 simulator with a five-stage, in-order pipeline and browser-based assembly execution controls.
- Implemented pipeline state visualization, register/memory inspectors, cycle history, instruction focus views, forwarding paths, bubbles, stalls, flushes, and retirement tracking.
- Added unit, component, and Playwright end-to-end tests covering CPU execution, pipeline hazards, forwarding, memory behavior, invalid programs, and UI workflows.

[Aviation Damage Risk Predictor](#) | *Python, pandas, scikit-learn, LightGBM, NumPy*

Mar – May 2026

- Built an imbalanced binary classification pipeline to predict indicated aircraft damage from 307K+ aviation incident records.
- Engineered 104 interpretable features from missingness patterns, date/time fields, high-cardinality categorical variables, text keywords, frequency encodings, and domain-specific risk interactions.
- Evaluated LightGBM variants with 5-fold stratified cross-validation, threshold tuning, and ablation testing; achieved 0.939 out-of-fold ROC-AUC and 0.860 balanced accuracy.

[Pacman AI Project Suite](#) | *Python, Search, Reinforcement Learning, Bayesian Networks*

Jan – May 2026

- Implemented AI agents across five Pacman projects covering graph search, adversarial search, value iteration, Q-learning, Bayesian network inference, and supervised classification.
- Applied these methods to maze navigation, multi-agent decision-making, reinforcement learning, probabilistic reasoning, and digit/Pacman behavior classification.

[Energy-Efficient Graphics API Benchmark](#) | *Vulkan, Direct3D 12, OpenGL, GPU-Z*

Feb – May 2026

- Benchmarked Vulkan, Direct3D 12, and OpenGL on an NVIDIA RTX 3070 Laptop GPU, collecting FPS, power, thermal, and GPU telemetry across repeated trials.
- Calculated FPS/W efficiency metrics, measuring Vulkan at 0.727 FPS/W versus 0.641 for OpenGL and 0.633 for Direct3D 12.

[QARDS – THE Quantum Card Game](#) | *Python, Qiskit, PySide6 (Qt)*

Sep – Oct 2025

- Built a digital quantum card game modeling superposition, entanglement, and interference as core gameplay mechanics.
- Implemented Qiskit gates, Grover oracle/diffusion mechanics, undo/reset actions, and statevector/measurement visualizations.
- Delivered both a CLI prototype and a GUI application, earning **2nd place** at the UT IBM Qiskit Fall Fest Hackathon (2025).